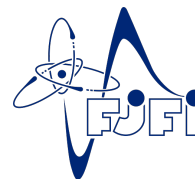




## 02YELMA Electricity and Magnetism



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### Course Description

This summer semester course for first-year bachelor students covers the basics of electricity and magnetism, providing a solid foundation in these fundamental concepts, which are essential for understanding more advanced topics in physics and engineering.

Each week, there will be four 50-minute lectures and two 50-minute exercise session. Lectures will cover theoretical concepts following the weekly schedule, while exercise sessions will focus on solving problems related to the lecture material. The course is conducted in English.

### Prerequisite(s)

Knowledge of basic differentiation, integration, and trigonometry would be useful but are not necessary.

**Credits: 6**

### Reference Textbooks

1. Griffiths, D. J., *Introduction to Electrodynamics*, Cambridge University Press, 3<sup>rd</sup> Edition, 1999.
2. Purcell, E. M., and Morin D. J. *Electricity and Magnetism*, Cambridge University Press, 3<sup>rd</sup> Edition, 2013.
3. Pandey, D. C., *Electricity and Magnetism*, Arihant Publications, 16<sup>th</sup> Edition, 2020.

### Course Objectives

1. To have a sound understanding of electricity and magnetism.
2. To be able to solve questions related to practical applications related to the subjects learned.

## Homeworks

Each week, students will be given homework on the topics covered in class. The purpose of homework is that students can work on their own and familiarize themselves with the subject. Homework will not be returned and will not be graded. Solutions to the homework assignments will be provided one week before the upcoming exam date.

## Exams

There will be **two midterm exams** and **one final exam**. Each exam will contain **four questions**, with each question worth **one point**. Students who obtain at least **50% of the total points** from the midterms will be assigned a Z grade and will be qualified for the final exam.

The final exam consists of a **written and an oral component**:

- Only students who score at least **50% on the written part** will proceed to the oral exam.
- The **final letter grade** is primarily determined by the written exam score. However:
  - If the oral exam score is **above 75%**, the final grade is increased by one level.
  - If the oral exam score is **below 25%**, the final grade is decreased by one level.
  - If the oral exam score is **between 25% and 75%**, the final grade remains the same.
- Attendance at the oral exam is **mandatory** for students who pass the written part.
- The structure of the oral part will be as follows: Each student will choose a random number from 1 to 16, corresponding to one of the sixteen topics listed at the end of this syllabus (please see “Topics for the Oral Part of the Final exam”). The students will then have 30 minutes to prepare a presentation on the chosen topic on a blank paper, followed by a 5 to 10-minute oral examination based on that topic. The students do not need to provide all the derivations (unless otherwise stated) but should be able to explain the main concepts and equations related to the topic.

The dates of the exams will be announced in the second week of the semester in KOS. The students are supposed to register for the exams in KOS. The students are required to bring their student ID cards to the exams.

## Letter Grade Distribution

|                             |                                |
|-----------------------------|--------------------------------|
| [3.6 – 4.0] : A = Excellent | [2.4 – 2.8) : D = Average      |
| [3.2 – 3.6) : B = Very Good | [2.0 – 2.4) : E = Sufficient   |
| [2.8 – 3.2) : C = Good      | [0.0 – 2.0) : F = Insufficient |

## Makeup Exams

If a student fails to obtain at least 50% of the total points from the midterm exams required to qualify for the final exam, they are eligible to take a makeup exam. The makeup exam will be scheduled before the official examination period. The structure and grading of the makeup exam will be identical to the midterms and will cover all the topics from the first two midterms. After taking the makeup exam, the student must achieve at least 50% of the total points from the combined score of the midterms and the makeup exam in order to be eligible to attend the final exam.

## **Retaking the Course**

If a student fails to obtain a Z grade for the first time, they must retake the course in the following academic year. The student is required to attend all classes again and retake all examinations. If the student fails this retake, they must obtain special permission from the study department to retake the course one final time.

If a student receives a Z grade but fails the final exam, the Z grade remains valid. In this case, the student is required to retake only the final exam in subsequent terms during examination period. If the student fails the final exam again, they must obtain special permission from the study department to retake the exam one final time.

## **Attendance and Absences**

- Attendance is strongly recommended but not obligatory. There is a strong correlation between attendance and performance in the exams, so students are encouraged to attend all lectures and exercise sessions.
- Students are responsible for all missed work, regardless of the reason for absence. It is also the absentee's responsibility to get all missing notes or materials.

## Tentative Course Outline

The weekly coverage might change as it depends on the progress of the class.

| Week    | Content                                                                                                                                                                                                                             |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1  | Vectors, vector operations, gradient, divergence, curl, and their fundamental theorems. Line, surface, and volume integrals. Curvilinear coordinates, Dirac delta function, vector field theory, potentials, and Helmholtz theorem. |
| Week 2  | Coulomb's law, superposition principle, electric fields from charge distributions, field lines, and Gauss's law.                                                                                                                    |
| Week 3  | Electric potential, Poisson's/Laplace's equation, work and energy in electrostatics.                                                                                                                                                |
| Week 4  | Boundary conditions, conductors, insulators, Faraday cage, surface charges, force on conductors, and capacitors.                                                                                                                    |
| Week 5  | Multipole expansion, dipoles, induced dipoles, polarization, bound charges, and their physical meaning.                                                                                                                             |
| Week 6  | Lorentz force, cyclotron motion, current density, continuity equation, Biot-Savart law, and Ampère's law.                                                                                                                           |
| Week 7  | Magnetic vector potential and its multipole expansion. Magnetic fields in matter, torque on magnetic dipoles, atomic orbits, magnetization, paramagnetism, diamagnetism.                                                            |
| Week 8  | Field of a magnetised object, bound currents and their physical interpretation, an introduction to ferromagnetism.                                                                                                                  |
| Week 9  | Conductivity, resistivity, Ohm's law, electrical circuits, electromotive force (emf), batteries, motional emf, and electromagnetic induction (Faraday's and Lenz's laws). Eddy currents.                                            |
| Week 10 | Inductance, RLC circuits, energy in magnetic fields, and Maxwell's equations.                                                                                                                                                       |
| Week 11 | Wave equation, sinusoidal waves, electromagnetic waves in vacuum, monochromatic plane waves, and Poynting's theorem.                                                                                                                |
| Week 12 | Einstein's postulates, time dilation, Lorentz contraction, Galilean and Lorentz transformations, spacetime structure, proper time/velocity.                                                                                         |
| Week 13 | Relativistic energy and momentum. Magnetism as a relativistic effect, field transformations, field tensor, and moving charge fields.                                                                                                |

## **Topics for the Oral Part of the Final Exam**

To be prepared...